DASHBOARD: -dashboard-overview.jpg

AgriVision

Agriculture Data Analytics

Demo

Season

2023–24

Crop filter

Rice

Overview | Production | Weather

Crop yield

4.2 t/ha

Illustrative data

Rainfall

720 mm

Seasonal total · demo

Soil moisture

34%

Sample reading

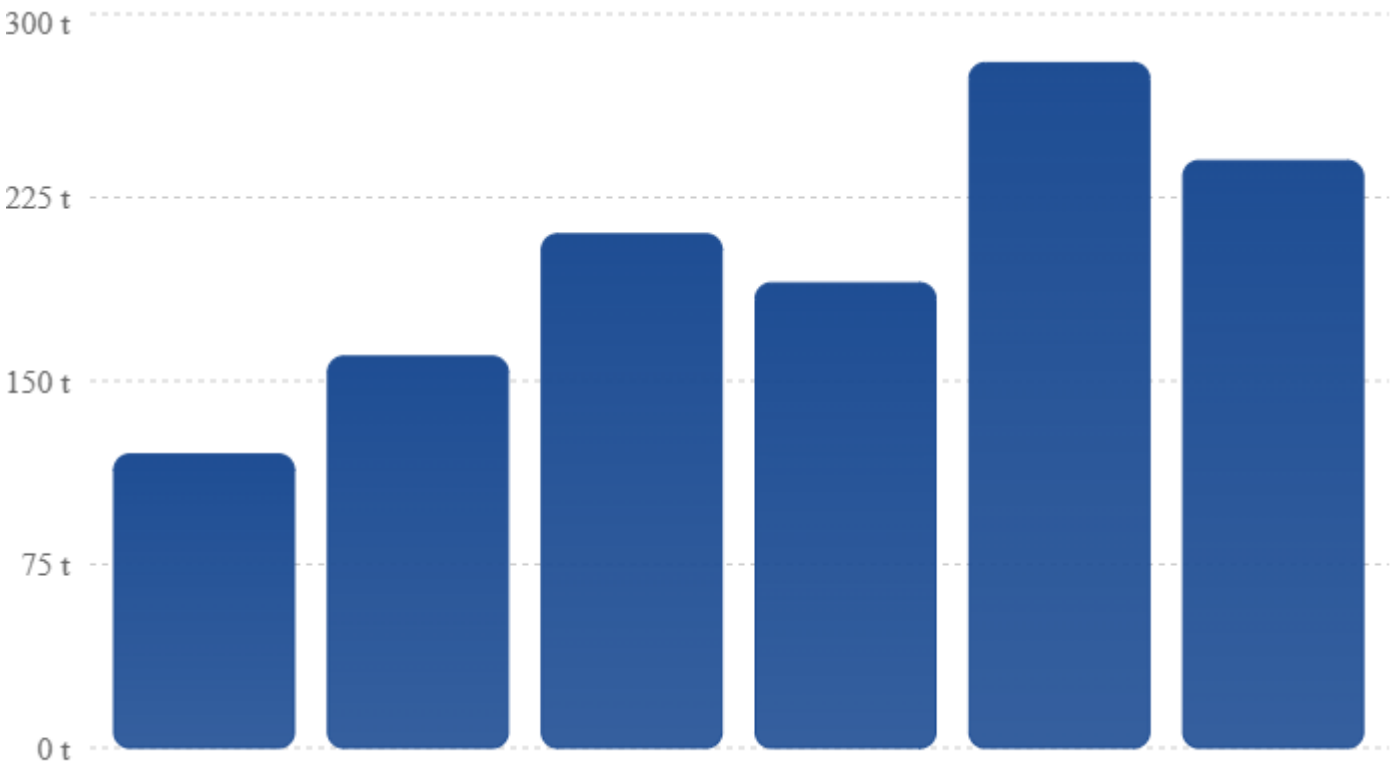
Farm area

1,250 ha

Example dataset

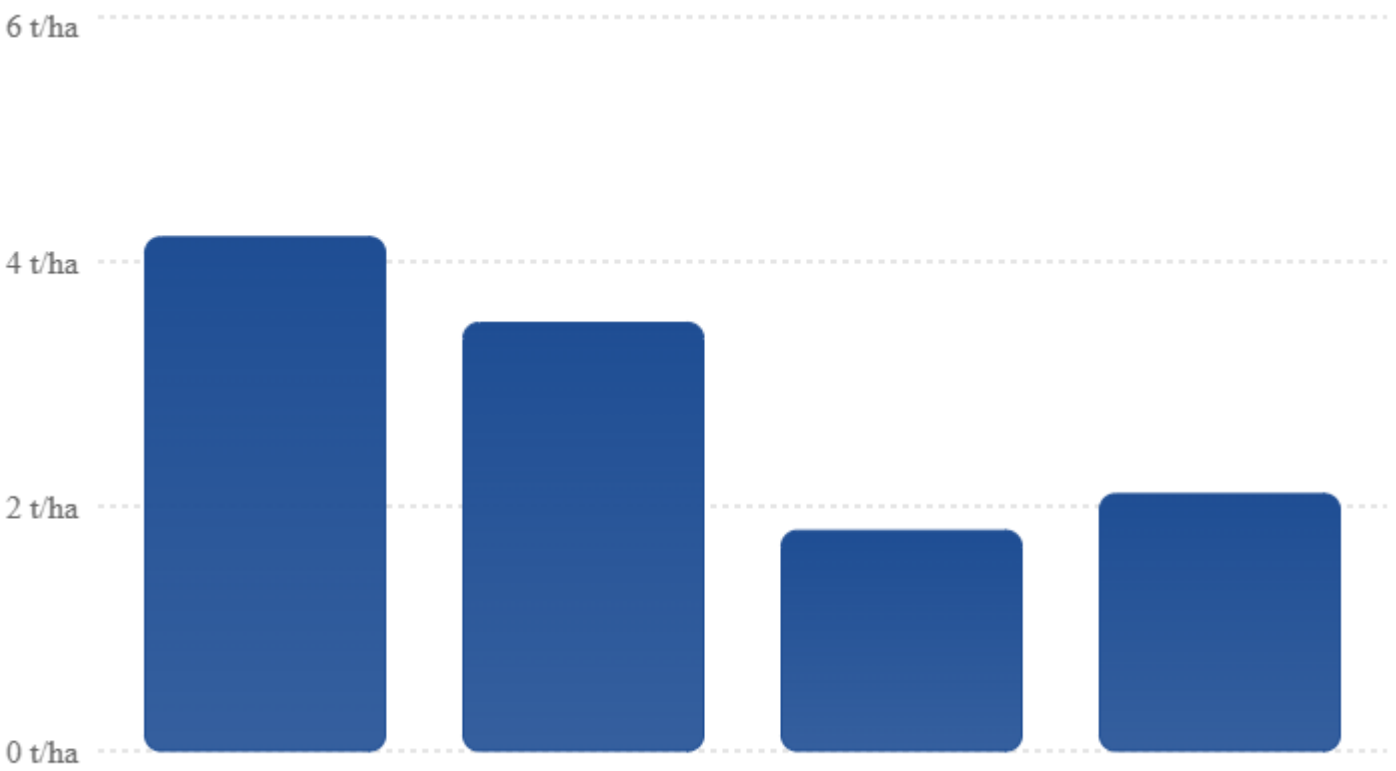
Monthly crop production

Illustrative production values · tonnes



Sample values for dashboard demonstration, not actual agricultural records.

Crop-wise yield comparison



Example yield values in tonnes per hectare.

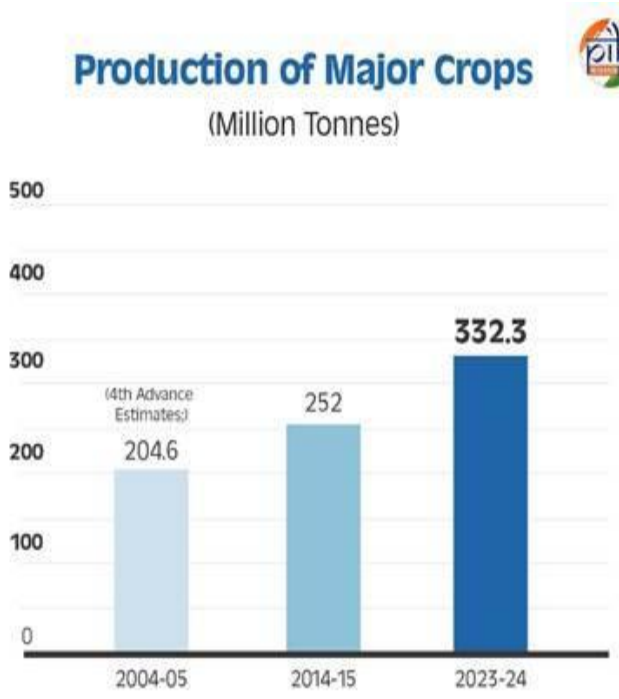
Types of data in graps:

1. Bar chart — Crop production comparison:

Compare the production of rice, wheat, maize, cotton, and sugarcane.

Data required: Crop name, state, production in tonnes.

Example insight: Which crop has the highest production?

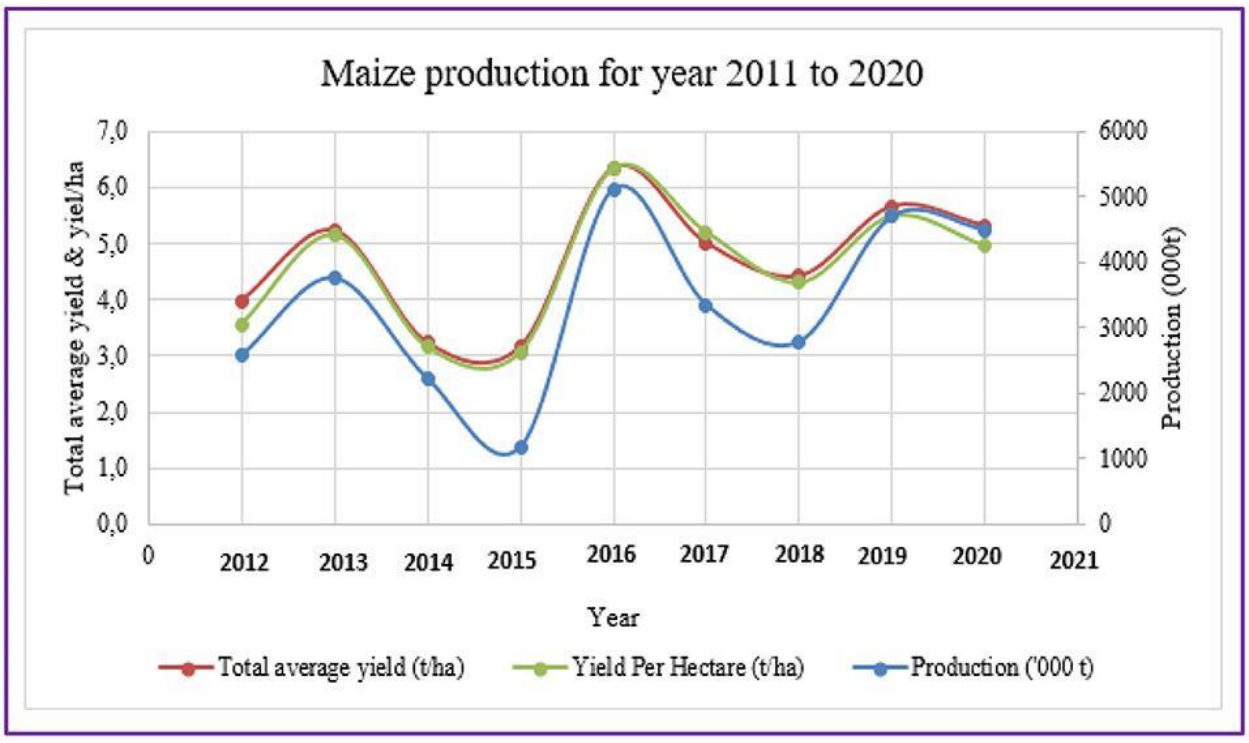


2.line chart: production over time

Show whether agricultural production increases or decreases over several years.

Data required: Year, crop name, annual production.

Example insight: How has rice production changed over time?

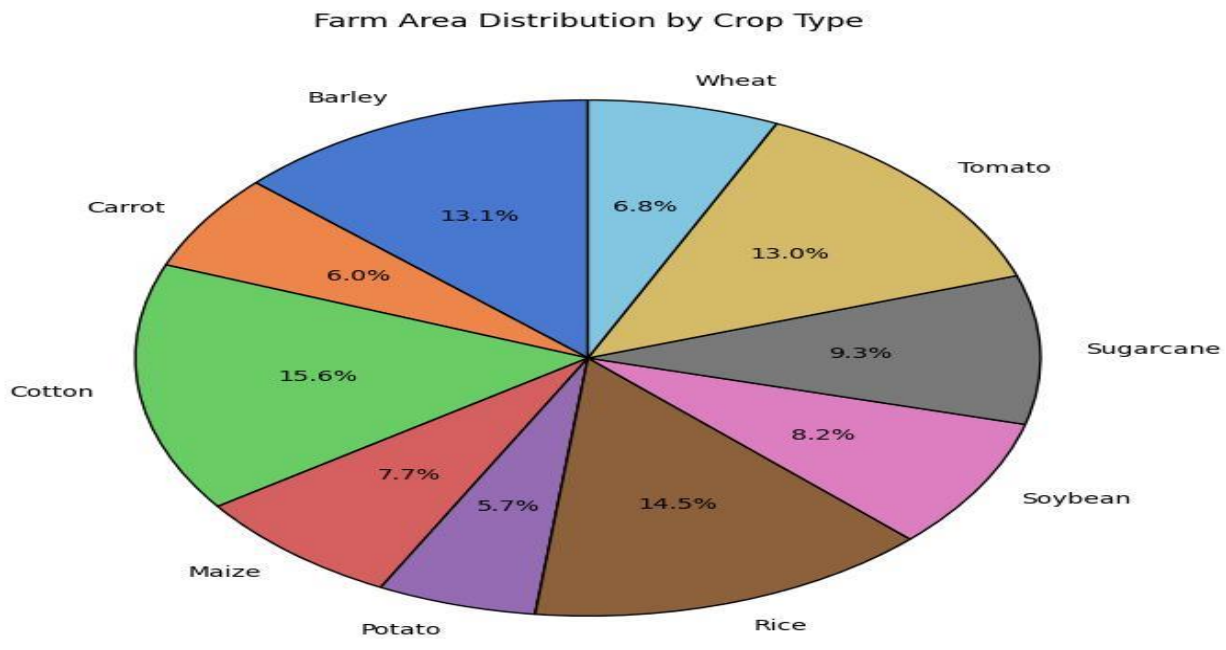


3. Pie chart — Crop production share

Display the proportion of total production contributed by each crop.

Data required: Crop name and production quantity.

Example insight: What percentage of the selected crops is rice?

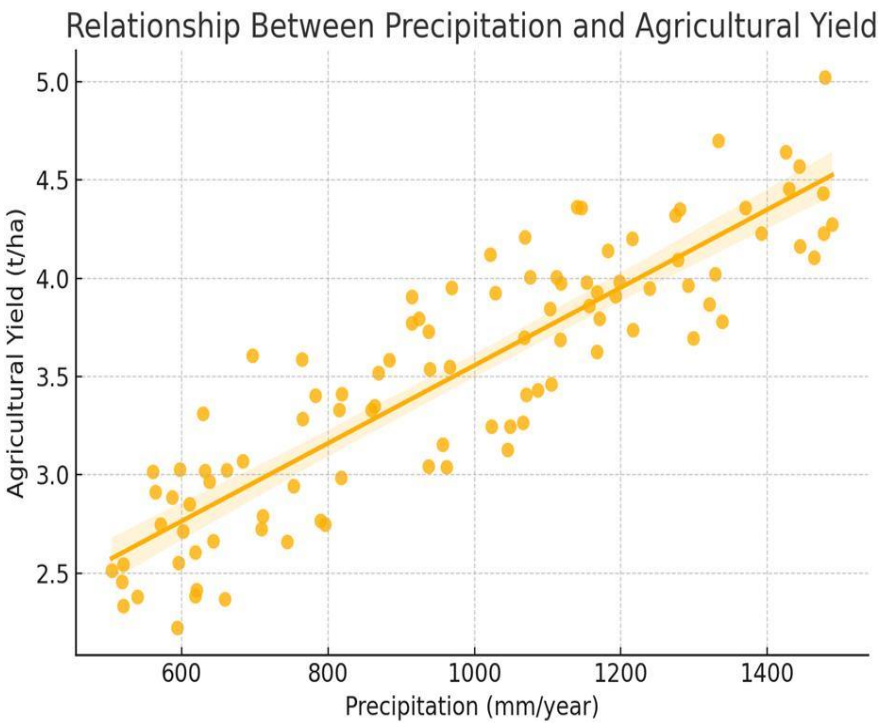


4. Scatter plot — Rainfall and crop yield

Examine the relationship between rainfall and crop yield.

Data required: Rainfall in mm and yield in tonnes per hectare.

Example insight: Do locations with different rainfall levels show different yields?

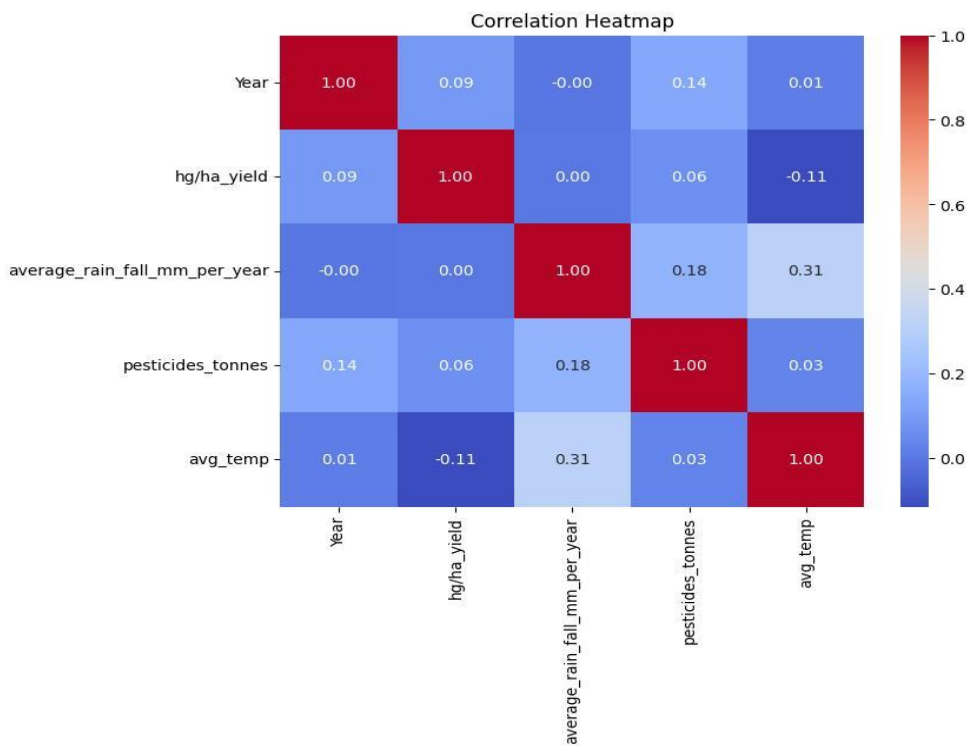


5. Heatmap — Agricultural patterns

Compare rainfall, temperature, soil moisture, or yield across regions and months.

Data required: Region, month, and a numeric measurement.

Example insight: Which months have higher rainfall?

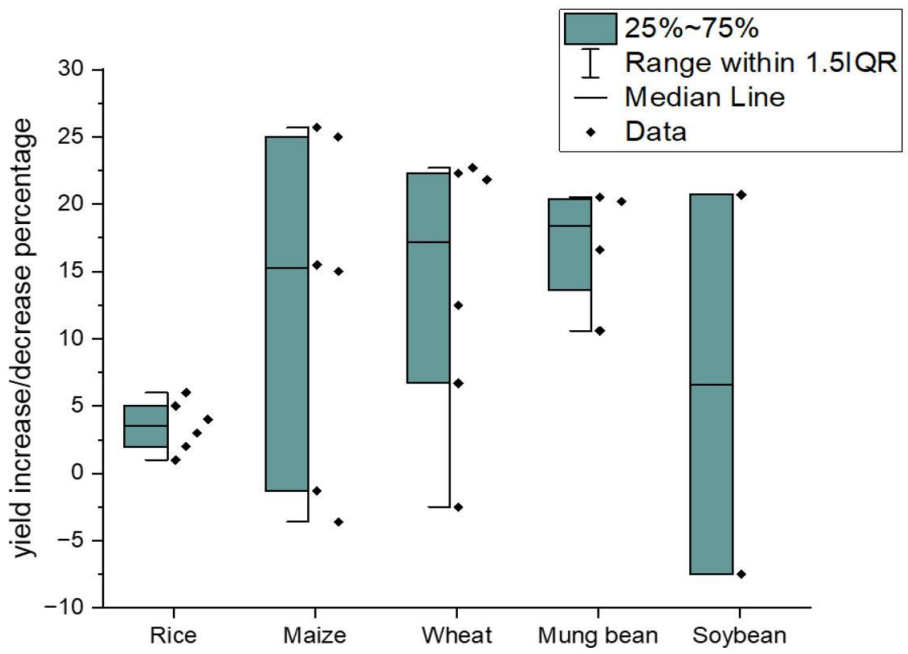


6. Box plot — Yield variation

Compare the spread of crop yields across states or seasons.

Data required: Crop, region, and yield measurements.

Example insight: Which crop has more variation in yield



Agricultural dataset with related data

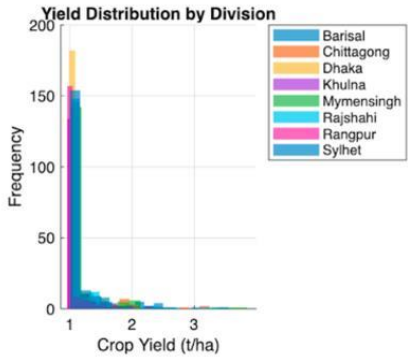
Your project needs a dataset containing columns that can be used to create graphs. You can start with the following sample data.

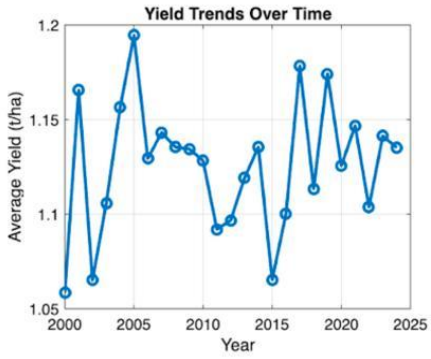
The values below are illustrative demo data, not verified agricultural statistics. Replace them with real data before drawing conclusions.

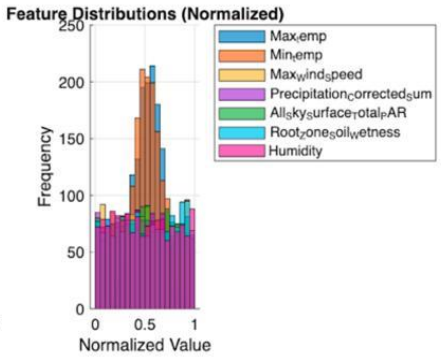
State	Crop	Year	Production (tonnes)	Area (hectares)	Rainfall (mm)	Yield (tonnes/ha)
Andhra Pradesh	Rice	2023	12000	3000	850	4.0
Telangana	Rice	2023	10000	2500	800	4.0
Punjab	Wheat	2023	15000	3000	600	5.0
Uttar Pradesh	Wheat	2023	14000	3500	650	4.0
Karnataka	Maize	2023	8000	2000	700	4.0
Maharashtra	Cotton	2023	6000	3000	750	2.0
Tamil Nadu	Rice	2024	13000	3000	900	4.33
Andhra Pradesh	Maize	2024	9000	2000	780	4.5
Telangana	Cotton	2024	6500	2500	720	2.6
Punjab	Wheat	2024	15500	3000	620	5.17

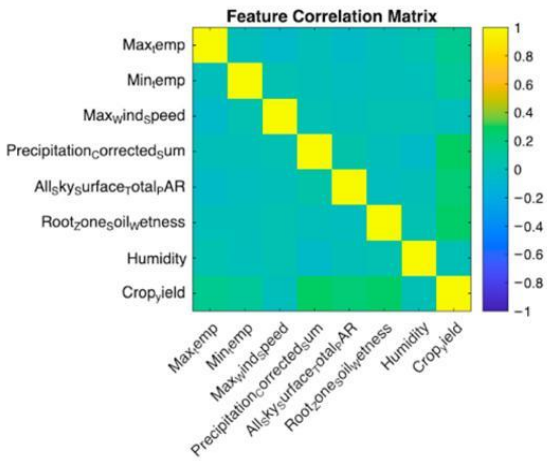
Graph	Required columns

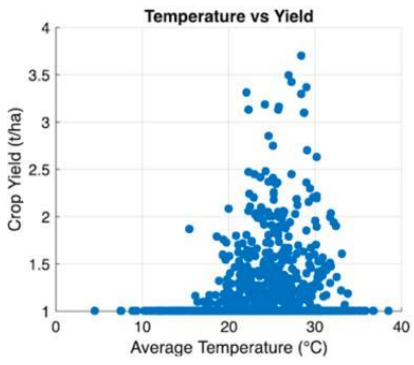
Bar chart	Crop, Production
Line chart	Year, Production
Pie chart	Crop, Production
Scatter plot	Rainfall, Yield
Heatmap	Month, State, Rainfall or Yield
Map	State, Production or Yield
Box plot	Crop, Yield
Stacked bar	State, Crop, Fertilizer type, Quantity

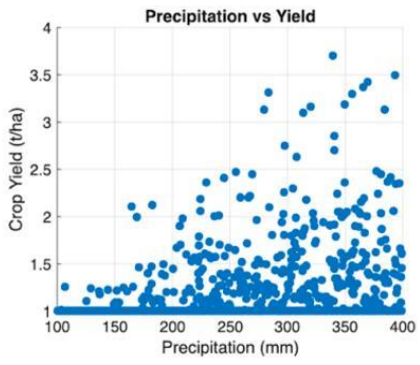












PROGRAM:

rogram

```
import streamlit as st
import pandas as pd
p
import plotly.express as px

# Page configuration
st.set_page_config(
    page_title="Agriculture Dashboard",
    page_icon="🌾",
    layout="wide"
)

# Title
st.title("🌾 Smart Agriculture Dashboard")
st.write("Agriculture Data Visualization using Streamlit")

# Load dataset
df = pd.read_csv("agriculture.csv")

# Sidebar
st.sidebar.header("🔍 Filters")

states = st.sidebar.multiselect(
```

```
    "Select State",
    df["State"].unique(),
    default=df["State"].unique()
)

crops = st.sidebar.multiselect(
    "Select Crop",
    df["Crop"].unique(),
    default=df["Crop"].unique()
)

seasons = st.sidebar.multiselect(
    "Select Season",
    df["Season"].unique(),
    default=df["Season"].unique()
)

# Filter data
filtered_df = df[
    (df["State"].isin(states)) &
    (df["Crop"].isin(crops)) &
    (df["Season"].isin(seasons))
]

# Dashboard metrics
col1, col2, col3, col4 = st.columns(4)

col1.metric(
    "🌱 Total Area",
    f"{filtered_df['Area_Hectares'].sum():,.0f} ha"
)

col2.metric(
    "🌾 Production",
    f"{filtered_df['Production_Tonnes'].sum():,.0f} tonnes"
```

```
)

col3.metric(
    "📈 Average Yield",
    f"{filtered_df['Yield_Tonnes_Ha'].mean():.2f}"
)

col4.metric(
    "🌧 Average Rainfall",
    f"{filtered_df['Rainfall_mm'].mean():.0f} mm"
)

st.divider()

# Chart 1
st.subheader("🌾 Production by Crop")

crop_data = (
    filtered_df
    .groupby("Crop")["Production_Tonnes"]
    .sum()
    .reset_index()
)

fig1 = px.bar(
    crop_data,
    x="Crop",
    y="Production_Tonnes",
    title="Production by Crop"
)

st.plotly_chart(fig1, use_container_width=True)

# Chart 2
st.subheader("🌧 Rainfall vs Yield")
```

```
fig2 = px.scatter(
    filtered_df,
    x="Rainfall_mm",
    y="Yield_Tonnes_Ha",
    color="Crop",
    hover_data=["State", "District"]
)

st.plotly_chart(fig2, use_container_width=True)

# Chart 3
st.subheader("🌱 Production by Soil Type")

soil_data = (
    filtered_df
    .groupby("Soil_Type")["Production_Tonnes"]
    .sum()
    .reset_index()
)

fig3 = px.pie(
    soil_data,
    names="Soil_Type",
    values="Production_Tonnes",
    title="Production Distribution by Soil Type"
)

st.plotly_chart(fig3, use_container_width=True)

# Dataset
st.subheader("📋 Agriculture Dataset")
st.dataframe(filtered_df, use_container_width=True)

# Download button
```

```
csv = filtered_df.to_csv(index=False)


st.download_button(
    "⬇️ Download Data",
    csv,
    "agriculture_data.csv",
    "text/csv"
)
```